Beyond the Factor of Safety

A CPT-Based Investigation of Liquefaction Triggering and Surface Manifestation Using the 2011 Christchurch Earthquake Case Histories

Nyarko Kennedy
Materials Laboratory, Ghana Highway Authority
July 2026 · Independent Geotechnical Research Project - Python implementation and analysis

This document condenses a full technical report into a short, self-contained summary for readers who need the essentials, the problem, the approach, what was found, and what it means without the full methodological detail.

Overview

WHAT THE PROJECT IS

This project builds a Python implementation of an established liquefaction-triggering procedure (Boulanger & Idriss, 2014) and applies it to six real CPT soundings from the 2011 Christchurch earthquake with one site for each observed severity level, from no visible ground damage to severe lateral spreading.

WHY IT MATTERS

Liquefaction assessment is central to geotechnical earthquake engineering as it determines whether saturated, loose soils will lose strength during shaking and cause ground failure. CPT data is one of the most widely used tools for this assessment because it gives a continuous profile of soil resistance with depth. Reproducing a published procedure end-to-end, on real case histories with a known outcome, is a strong way to demonstrate genuine understanding of the method rather than just its formulas.

WHAT WAS ACTUALLY TESTED

Two questions were investigated together:

- Does the calculated Factor of Safety (FS) against liquefaction line up with what was actually observed at the ground surface in Christchurch?
- Does correcting the CPT data for thin soil layers, a known limitation of the cone penetration test change that picture?

HEADLINE RESULT

All six sites, regardless of how much surface damage they actually showed, came out with FS below 1.0 somewhere in the profile including the site with no observed manifestation at all. Also the calculated FS did not rank the sites in the same order as their real-world severity. This is the central, deliberately counter-intuitive finding the report is built around hence the title "Beyond the Factor of Safety."

Introduction

Background

Earthquake-induced liquefaction happens when loose, saturated soil loses strength under cyclic shaking, generating excess pore pressure and causing ground deformation. The 2011 Christchurch earthquake is a well-documented case history for this, because it produced a wide range of surface effects from nothing visible to severe lateral spreading across sites that were also extensively investigated with CPT. This combination of measured subsurface data and known field outcomes makes it possible to test whether an analytical procedure's predictions match reality.

The CPT-based method used here, developed by Boulanger and Idriss (2014), compares the cyclic resistance the soil can offer (CRR) against the cyclic demand imposed by the earthquake (CSR). If their ratio gives the Factor of Safety below 1.0, it signals a susceptibility to triggering at that depth. A separate, known complication is that, the CPT cone tends to blur the true resistance of thin soil layers because of how it senses the surrounding ground. Boulanger and DeJong (2018) proposed a correction (inverse filtering) for this.

Aim

The aim of this project was to build a working Python implementation of the Boulanger & Idriss (2014) procedure, apply it to six real Christchurch case histories, and examine whether correcting for thin-layer effects changes the predicted factor of safety and whether either version of the prediction tracks the severity that was actually observed on the ground.

Objectives

- Implement the full CPT-based triggering procedure in Python, from raw CPT data through to Factor of Safety.
- Process six real Christchurch case histories using Python data-handling tools.
- Calculate soil behaviour type index, fines content, cyclic resistance ratio (CRR), cyclic stress ratio (CSR), and FS for each site.
- Run the analysis on both the as-measured CPT data and the thin-layer-corrected data, and compare the two.
- Compare predicted FS against each site's real observed manifestation class (no damage → severe lateral spreading).
- Critically examine any cases where the analytical prediction and the observed field behaviour disagree.

Note: The six sites were deliberately treated as individual case studies, not a statistical sample. The goal was to demonstrate the method and probe its behaviour, not to statistically validate it.

Data and Method

Where the data came from

The CPT soundings and earthquake parameters came from the Canterbury CPT-based liquefaction case-history dataset (Geyin et al., 2021), which packages real CPT measurements (tip resistance, sleeve friction, pore pressure), groundwater depth, earthquake magnitude, peak ground acceleration, and a manually assigned liquefaction manifestation class for each site.

How the six sites were chosen and why

Rather than picking sites arbitrarily, a filtering process was applied first, to make sure every candidate site was actually suitable for this kind of assessment.

- Sites with an unknown manifestation classification were dropped.
- Sites where the CPT's pre-drilled section extended below the groundwater table were excluded, since that could distort the stress profile.
- Sites shaken more weakly than 0.075g PGA were excluded since they were too weak to be a meaningful liquefaction test.
- Only soundings reaching at least 6 m depth were kept, to ensure enough profile to work with.

From the sites that passed all four checks, one CPT sounding was randomly selected (fixed random seed, for reproducibility) from each of the six manifestation classes 0 (none) through 5 (severe lateral spreading) giving one matched case study per severity level.

The calculation pipeline

For each site, the Python implementation worked through the standard chain of calculations shown below starting from the corrected CPT resistance and ending at the Factor of Safety at every depth. Soil layers unlikely to be liquefiable (Ic > 2.6) were excluded before taking the minimum FS for each site, following standard practice.

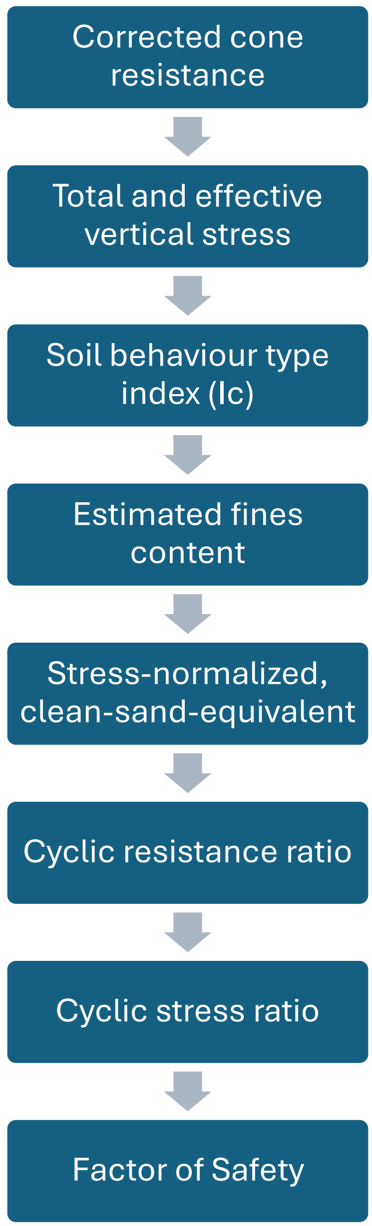

Figure 1. An eight-stage calculation applied at each depth across all sites.

Why the thin-layer comparison was done

The dataset provides both the as-measured CPT resistance and a version already corrected for thin-layer/transition-zone effects using Boulanger & DeJong's (2018) inverse-filtering method. Running the full pipeline on both versions, side by side, made it possible to isolate how much of the final result depends on this CPT data-processing choice, versus the underlying assessment method itself.

A few implementation details worth knowing

- A minimum effective stress of 1 kPa was enforced to avoid divide-by-zero errors right at the ground surface (depth = 0).
- The measured and thin-layer-corrected profiles were aligned to a shared starting depth, so the two versions were compared over exactly the same depth range.
- Normalization was solved iteratively (starting guess, then refined until the change was under a set tolerance), since the stress-normalization exponent depends on the very resistance value it's used to calculate.

Results

Minimum factor of safety by site

Every one of the six sites returned a minimum FS below 1.0 including the Class 0 site, where no liquefaction was actually observed at the surface. The table below summarizes the results for both the measured and thin-layer-corrected CPT data.

Class	Observed Manifestation	Min. FS (Measured)	Min. FS (Corrected)
0	No manifestation	0.324	0.349
1	Minor	0.310	0.334
2	Moderate	0.259	0.264
3	Severe	0.269	0.347
4	Lateral spreading	0.424	0.450
5	Severe lateral spreading	0.327	0.317

Table 1 Shows manifestation classes with corresponding minimum FS values for both measured and corrected.

The prediction does not track observed severity.

The clearest picture of the central finding is this trend line. If the method worked as a direct severity predictor, FS should fall steadily from Class 0 to Class 5. It doesn't.

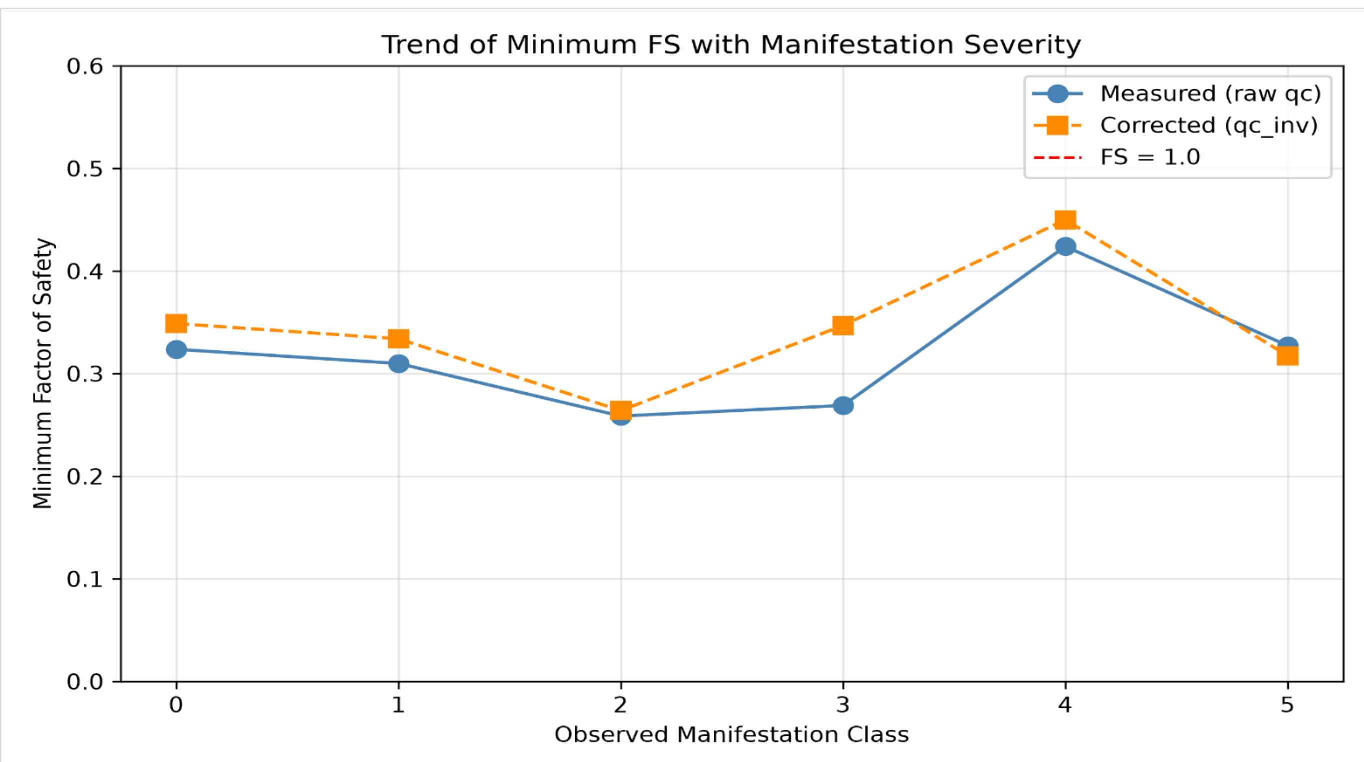

Figure 2 Trend of minimum FS across observed manifestation classes (measured vs. thin-layer-corrected).

The lowest FS (0.259) occurred at Class 2 (moderate manifestation) not at Class 5. The highest FS (0.424 / 0.450) occurred at Class 4 (lateral spreading) despite that site actually spreading laterally. And Class 0 (no damage) produced almost the same FS (0.324) as Class 5 (severe lateral spreading, 0.327).

Effect of the thin-layer correction

Correcting for thin layers raised the minimum FS at five of the six sites (Classes 0 - 4), with the biggest jump at Class 3 (0.269 → 0.347). Class 5 was the exception, dropping slightly (0.327 → 0.317). Importantly, the correction never changed a site's status relative to FS = 1.0. It shifted the numbers but not the overall triggering conclusion at any site.

Depth profiles

Looking at FS with depth (rather than just the single minimum value) shows where the weak zones actually sit in each profile, and how thick or thin they are. Low-FS zones appear at different depths and with different extents across the six sites, information the single minimum-FS number alone discards.

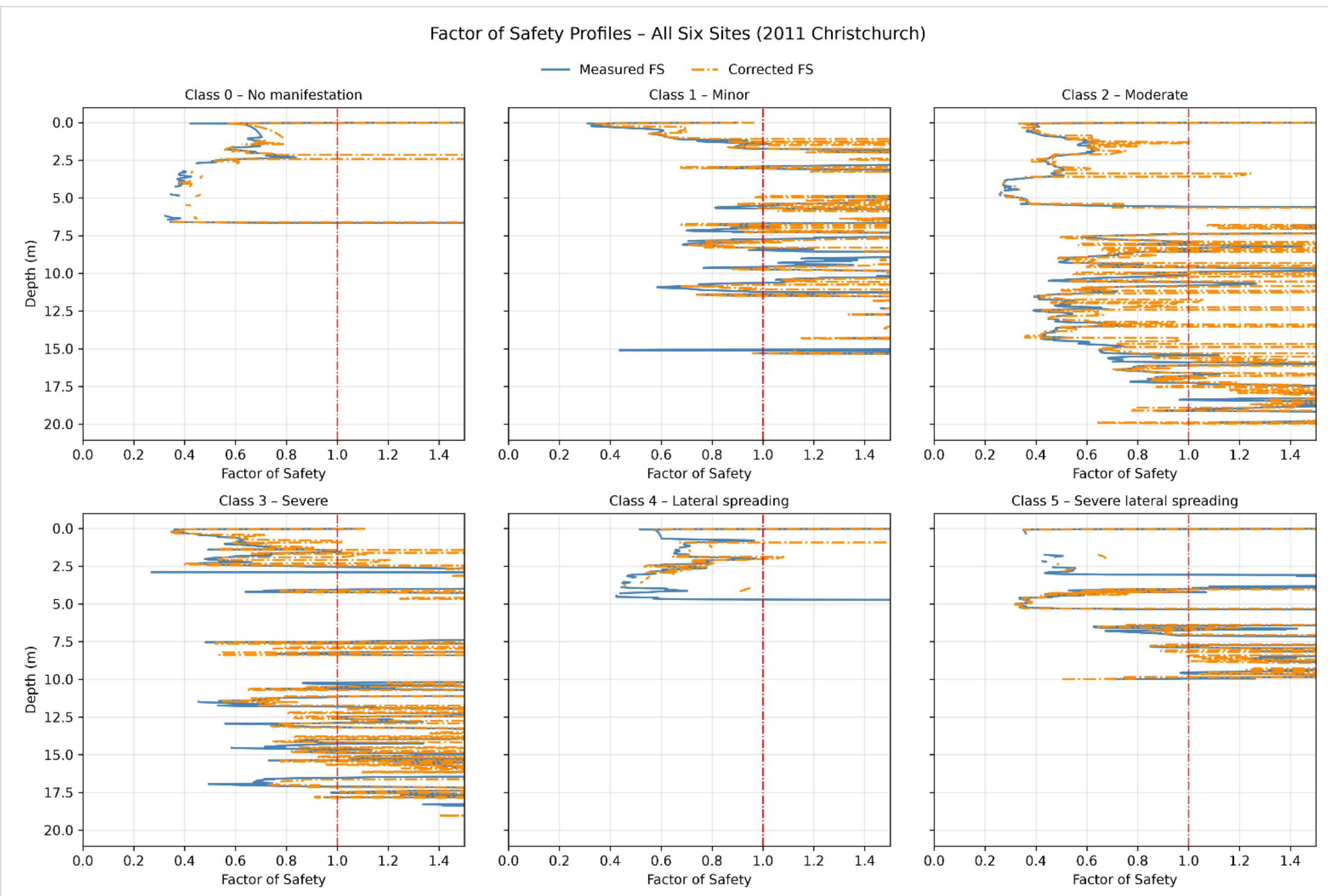

Figure 3. Factor of safety profiles versus depth for all six selected CPT sites (measured data).

Comparing the Class 0 site (no manifestation) directly against the Class 5 site (severe lateral spreading) makes the disconnect concrete as both dip below FS = 1.0, but the ground responded in completely different ways.

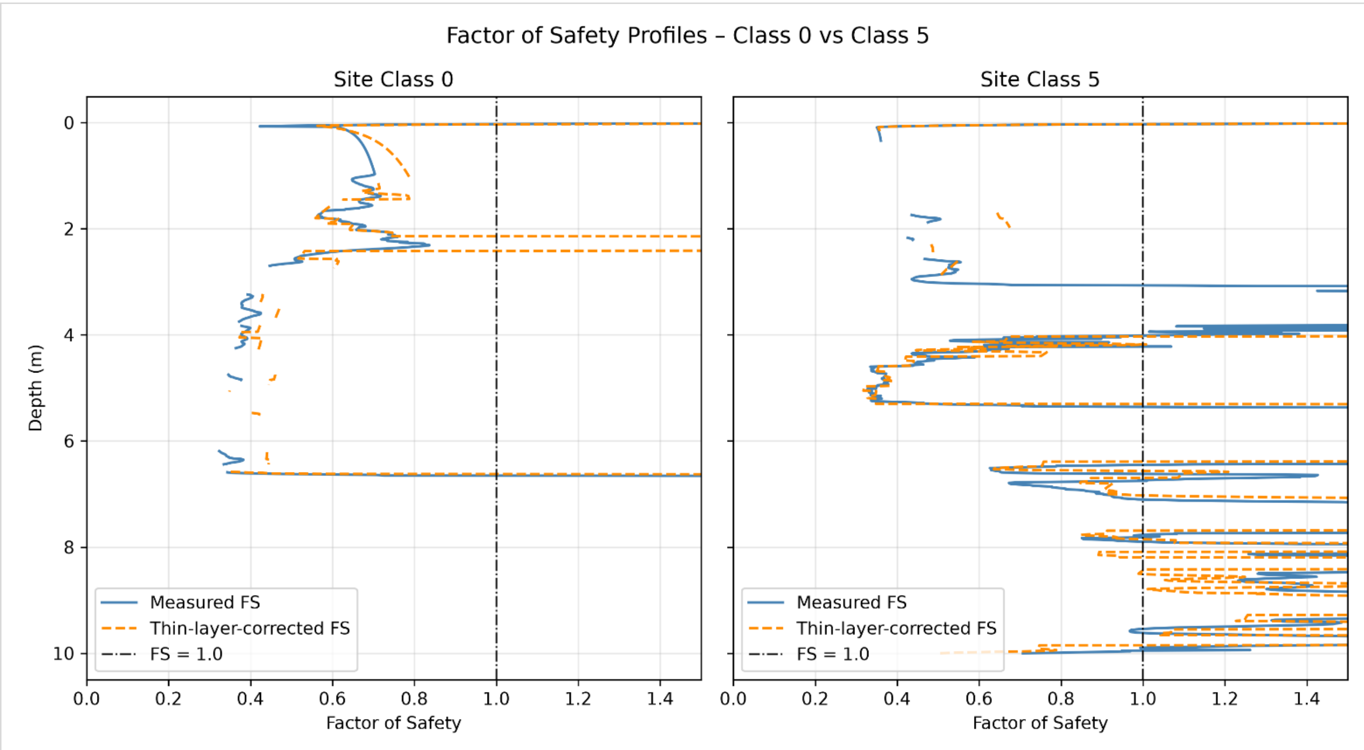

Figure 4. Factor of safety profiles versus depth: Class 0 (no manifestation) compared with Class 5 (severe lateral spreading).

Discussion and Key Findings

The results point to one central conclusion, a low calculated Factor of Safety tells you a soil layer is susceptible to triggering but it does not tell you how that will show up, or whether it will show up at all, at the ground surface.

- Triggering vs. manifestation are different questions. FS is a one-dimensional, depth-specific comparison of resistance and demand. Surface manifestation depends on much more like the thickness and continuity of the weak layer, whether a non-liquefiable crust caps it, drainage paths, groundwater conditions, and site geometry (slopes, free faces, nearby waterways for lateral spreading in particular).
- A low minimum FS can come from one localized weak layer that never develops into visible surface damage.
- Lateral spreading (Class 4) is driven partly by geomorphology, slopes, free faces, channels that a 1-D FS calculation simply cannot see, which plausibly explains why that site had a comparatively high FS despite visible spreading.
- Thin-layer correction matters for the numbers but is not a fix for the triggering-vs-manifestation gap. It changed FS magnitude, sometimes substantially, but never realigned the sites into the observed severity order.

Taken together, these findings support a fairly nuanced, practice-relevant conclusion that CPT-based FS is a genuinely useful screening tool for identifying susceptible layers, but treating it as a direct severity forecast as it is sometimes informally used is not supported by these six case histories.

Limitations

- Small sample: One site per manifestation class (six total) is not statistically representative. The conclusions are case-study observations, not a validated statistical relationship.
- Minimum-FS focus: The analysis leans on a single worst-case value per site, which does not capture layer thickness, continuity, or how much of the profile is actually susceptible.
- Triggering only, not deformation: The study evaluates whether liquefaction is triggered, not how much settlement or lateral spreading would actually result. The mechanisms behind Class 4/5 manifestations go beyond what a CRR–CSR comparison can model.
- Empirical correlations carry their own uncertainty: The soil behaviour, fines-content, and resistance correlations used were developed from specific calibration datasets and inherently carry uncertainty when applied to any individual site.

Conclusions and Recommendations

Conclusions

A complete, working Python implementation of the Boulanger & Idriss (2014) CPT liquefaction-triggering procedure was built and applied to six real Christchurch case histories, run against both measured and thin-layer-corrected CPT data. The procedure correctly flagged susceptibility (FS < 1.0) at every site, but the minimum FS did not reproduce the actual ranking of surface damage severity most notably, the no-manifestation site and the severe lateral-spreading site returned almost identical minimum FS values. The thin-layer correction changed FS magnitudes, sometimes substantially, but never changed which side of FS = 1.0 a site fell on. The project's core value is demonstrating both the mechanics of the method and its practical limits when checked against real field outcomes.

Recommended future work

- Expand to more case histories per class for a statistically meaningful comparison.
- Look beyond minimum FS to the thickness/continuity of susceptible layers.
- Extend into post-liquefaction deformation (settlement, lateral spreading) analysis, not just triggering.
- Bring in site geology/geomorphology (slope, free faces, waterways) to help explain manifestation severity.
- Compare against other CPT-based triggering procedures.
- Turn the Python workflow into a reusable, automated tool for batch-processing many CPT sites.
- Add uncertainty analysis across the CPT measurements and empirical correlations used.

Full methodology, equations, all figures, and the complete reference list are available in the original technical report.